7	(a)	(i)	$F_A = \frac{GM_1M_2}{(R_1+R_2)^2}$ (correct expression)	1
		(ii)	$F_B = M_2 R_2 \omega^2$ (correct expression) OR same expression as (a)(i)	1
	(b)		$\omega = 2\pi / T = 2\pi / (1.26 \times 10^8)$ (correct expression + substitution) $= 4.99 \times 10^{-8}$ rad s^{-1} (correct final answer, 3 s.f.)	1 1
	(c)	(i)	Both stars experience the same centripetal force because they experience the same gravitational force (which provides the centripetal force) Thus, $M_2 R_2 \omega^2 = M_1 R_1 \omega^2$ $\Rightarrow \frac{M_2}{M_1} = \frac{R_1}{R_2}$ (shown)	1 1
		(ii)	$\frac{1}{3} = \frac{R_1}{3.2 \times 10^{11} - R_1}$ (correct substitution) $\Rightarrow 3R_1 = 3.2 \times 10^{11} - R_1$ $\Rightarrow R_1 = 8.0 \times 10^{10}$ m and $R_2 = 2.4 \times 10^{11}$ m (correct final answers, 2/3 s.f.)	1 1
	(d)		Both stars are orbiting in a plane containing Earth's line of sight. So at two instances in one oscillation, one star will block the other star from our view. That is the reason for the reduction in intensity. AND EITHER (Addresses difference in dip) One star is brighter than the other star. That is why the reduction in intensity is different. OR (Addresses why shape of dip is V) Both stars are of similar size. That is why the dip in intensity is V-shaped (as opposed to U-shaped)	1 1 1
	(e)		A point charge may be positive or negative / the electric potential can either be positive or negative So if it is positive, when distance increases, the electric potential decreases. This is different from gravitational potential. If the charge is negative, when distance increases, the electric potential increases, which is the same as gravitational potential.	1 1
	(f)	(i)	$U = \frac{1}{4\pi\epsilon_0}\frac{Q_1Q_2}{r} = \frac{1}{4\pi\epsilon_0}\frac{(3\times1.6\times10^{-19})(1.6\times10^{-19})}{(2.3+1.2)(10^{-15})}$ (correct expression) (correct substitution) $= 1.97 \times 10^{-13}$ J (correct answer, 2/3 s.f.)	1 1 1

		(ii)	For the proton to travel from infinity and just make contact with the lithium nucleus, it must have possessed an initial kinetic energy of 1.97×10^{-13} J,	1
			which means it had an initial speed of about 5% the speed of light.	1
			Thus, the kinetic energy is of a high amount.	
	(g)		Work done on e^- by p.d. between plates = Gain in KE of e^-	1
			$\Rightarrow \quad q\,\Delta V = \frac{1}{2} m v_Y^2 - \frac{1}{2} m v_X^2$ $\quad \boxed{2.04975 \times 10^{-17} \text{ J}}$	
			$\Rightarrow \quad (1.6 \times 10^{-19})\ \Delta V = \frac{1}{2}(9.11 \times 10^{-31})\,[(9 \times 10^6)^2 - (6 \times 10^6)^2]$	
			$\Rightarrow \quad \Delta V = 128.109$ V (correct expression + substitution)	1
			Since $E = \Delta V / s$, we have	
			$s = \frac{128.109}{1.4 \times 10^5} = 9.15 \times 10^{-4}$ m (correct answer, 2/3 s.f)	1
8	(a)	(i)	Wavelength = speed × time for 1 cycle	1